Do Multi-Hop RAG Evaluations Measure Agentic Behavior? A Failure-Mode Audit

Anonymous ACL submission

Abstract

Modern multi-hop retrieval-augmented generation (RAG) systems decide *what* to retrieve, *when* to stop, and *how* to recover, but their evaluations may not measure this control behavior. We audit a documented, purposive corpus of 32 iterative, graph-based, and agentic multi-hop RAG papers across 10 outcome, process, and cost signals. Repeat coding of a seeded, stratified 22% sample gives 97.1% agreement (pooled Cohen's $\kappa = 0.93$). Within this corpus, 97% report final-answer accuracy, 13% joint evidence completeness, and 3% retrieval-control quality; no iterative/agentic paper reports recovery (0/15), and no tool-agent reports tool-error rate (0/5). We turn recurring failure modes into an operational table linking trigger, observable signal, consequence, recovery, and detecting metric, plus a practitioner checklist. A retrieval-only illustration over all 22,398 multi-support questions in the official 2Wiki, MuSiQue, and HotpotQA development splits shows joint evidence completeness 8–31 points below aggregate recall@20. Rather than another architecture survey, this is a transparent audit showing that systems with iterative or agentic components are usually scored on endpoints, not their control behavior.

1 Introduction

Multi-hop question answering (QA) requires assembling several interdependent facts into a reasoning chain. RAG has responded with a progression of designs: dense and late-interaction retrievers, graph systems that traverse entity relations, iterative retrieve-then-reason loops, and tool-using agents that plan their own retrieval. A recurring narrative motivates this progression: a set of *failure modes* (silent retrieval failures, context fragmentation, broken graph paths, cascading errors from a wrong hop) that each new design claims to mitigate through more agentic control over retrieval.

These failure modes are behavioral: they concern *how* a system retrieves, decides to stop, and recovers, not merely whether the final answer string matches. If a system is agentic, its evaluation should expose process quality. We ask a simple, falsifiable question: **do multi-hop RAG evaluations measure the retrieval-control behavior that iterative and agentic designs introduce?**

We answer with a quantitative audit rather than another narrative survey. We assemble a documented, purposive corpus of 32 iterative, graph-based, and agentic multi-hop RAG papers through a documented protocol and code each for whether it reports a set of outcome, agent-quality, and cost signals (§2). We then (i) recast the standard failure taxonomy as an *operational* table that maps each trigger to the observable signal and metric that would catch it (§3); (ii) quantify how often each signal is actually reported (§4); and (iii) distill a practitioner checklist (§6).

Recent surveys catalog RAG *methods* and reasoning strategies (Gao et al., 2023; Li et al., 2025; Deng et al., 2026). Our focus differs: we audit *evaluation practice*. Our contributions are:

- A transparent coding protocol and appendix matrix auditing multi-hop RAG evaluation (32 papers, 10 dimensions, with a repeat-coding consistency audit on 22% of the sample).
- An operational failure-mode table (trigger → signal → consequence → recovery → metric) that turns descriptive limitations into measurable quantities.
- Quantified reporting gaps showing process-level signals are rarely reported in this sample, and a checklist tying task properties to the retrieval strategy and the metrics to report.
- A full-split retrieval illustration (2Wiki, MuSiQue, HotpotQA; 22,398 questions) showing that the omitted evidence-completeness metric exposes an 8–31 point gap, with larger gaps as the number of

required passages grows.

2 Method

Corpus construction. We searched arXiv and the ACL/EMNLP/NAACL and NeurIPS/ICLR/SIGIR proceedings (2020–2026) using queries combining "multi-hop" with {RAG, retrieval-augmented, iterative retrieval, agentic retrieval, graph RAG, knowledge-graph QA}. We constructed a purposive sample covering four design families and several influential adjacent systems whose evaluations include retrieval-like tool use or multi-step control. The corpus contains iterative decomposition (10), tool-using agents (5), graph-based reasoning (15), and iterative dense retrieval (2). It is intended to characterize reporting practice across these families, not to estimate architecture prevalence in the full literature. Appendix A gives the full list, source links, and per-paper codes.

Coding scheme. For each paper we read the evaluation section and code, as $1/0/?$ (reported / not reported / unclear), the dimensions in Table 1: two *outcome* signals, five *agent-quality* signals, and three *cost* signals. We coded conservatively: a dimension is 1 only when the paper reports a corresponding quantitative measure. To keep judgments defensible we pre-specified conventions before final reconciliation, e.g.: aggregate recall@k that credits *partial* support is *not* evidence completeness (A1 requires a joint/all-support metric such as joint passage-set EM); "retrieval frequency" (fraction of steps that trigger retrieval) counts as retrieval use/calls (C1); offline indexing time alone does not count as latency (C3). For trajectory correctness (A4), a quantified manual error analysis over a stated sample counts, while isolated examples do not.

Consistency check. With seed 20260803 we sampled seven papers, stratified as two iterative, one agentic, three graph, and one dense system. A source-verification pass re-coded all 70 cells before reconciliation. Raw agreement was 68/70 (97.1%) and pooled Cohen's $\kappa = 0.93$. The two source-resolved disagreements corrected MDR's trajectory code and HippoRAG 2's latency code. Appendix A reports both code matrices and the sample rule.

Table 1: Coding dimensions. Agent-quality signals (A) are the behaviors that distinguish an agent from a static pipeline.

Group	Dimension
Outcome (S)	answer EM/F1/acc; retrieval recall@k
Agent (A)	evidence completeness (joint support) retrieval-control decision quality failed-retrieval recovery trajectory / path correctness tool-error rate
Cost (C)	retrieval use/calls; tokens/cost; latency

3 Failure Modes as Observable Signals

The literature's failure modes are usually stated descriptively. Table 2 recasts them operationally: each trigger produces an observable signal, has a downstream consequence, and admits a recovery action. Crucially, each trigger maps to a metric that would *detect* it. The rightmost column is what we audit for in §4; every row names the measurement that operationalizes it.

4 Findings: The Reporting Gap

Table 3 and Figure 1 report the fraction of the 32 systems that measure each dimension. The pattern is stark. Final-answer accuracy is nearly universal (97%), and retrieval recall is common (44%). But the agent-quality signals that the failure modes in Table 2 call for are rarely measured: evidence completeness 13%, retrieval-control quality 3%, and trajectory correctness 16%. Using relevant family denominators, only 1/15 iterative or agentic papers reports control quality, none reports failed-retrieval recovery, and none of the five explicitly tool-using agents reports tool-error rate. Even the near misses quantify retrieval *use*, such as Toolformer's call frequency and FrugalRAG's search count/cost (Schick et al., 2023; Java et al., 2025), rather than failure or recovery rates. Cost signals are reported unevenly (calls 38%, tokens 41%, latency 41%), and disproportionately by the subset of papers whose contribution *is* efficiency.

Where the exceptions are. The few agent-quality measurements are instructive. *Evidence completeness* is measured by only four systems, all via a joint/all-support metric: MDR's joint two-passage recall (Xiong et al., 2021), HippoRAG's All-Recall@k (Gutiérrez et al., 2024), Beam Retrieval's joint gold-passage-set EM (Zhang et al., 2024), and LinearRAG's all-necessary-information

Table 2: Operational failure-mode table for multi-hop RAG. Each descriptive failure becomes a measurable quantity (rightmost column). A/S/C refer to the coding dimensions in Table 1.

Failure trigger	Observable signal	Downstream consequence	Recovery action	Detecting metric
Embedding bottleneck (silent retrieval failure)	gold passage ranked low / missed despite plausible top-k	incomplete evidence; confident wrong answer	query reformulation; multi-vector / late interaction	evidence recall@k; joint support recall (S2/A1)
Context fragmentation	a required fact split across chunks; one piece absent	broken reasoning chain	semantic chunking; neighbor expansion	evidence completeness (A1)
Premature / late stopping (iterative)	loop halts before all hops, or over-retrieves	missing evidence or wasted budget	sufficiency check; adaptive stop	control-decision quality (A2); retrieval calls (C1)
Failed / wrong intermediate hop	a hop returns wrong or empty evidence	later hops condition on bad evidence (cascade)	backtrack; re-query; verification	recovery rate (A3); trajectory correctness (A4)
Shortcut reasoning (right answer, wrong path)	answer correct but support unused / spurious	brittle, non-generalizing behavior	path supervision; distractor stress test	trajectory correctness (A4)
Tool / action-call error (agentic)	malformed query or failed tool call	dropped or corrupted step	error handling; retries	tool-error rate (A5)
Graph construction gaps / missing edges	reasoning path halts at an absent edge	unreachable evidence	edge completion; hybrid fallback	support recall on reachable golds (S2/A1)

Table 3: Corpus-wide reporting rate per dimension (fixed denominator 32). Family-relevant denominators for control, recovery, and tool errors are reported in the text.

Group	Dimension	Reported
Outcome	answer EM/F1/acc	31/32 (97%)
	retrieval recall@k	14/32 (44%)
Agent	evidence completeness	4/32 (13%)
	retrieval-control quality	1/32 (3%)
	failed-retrieval recovery	**0/32 (0%)**
	trajectory correctness	5/32 (16%)
	tool-error rate	**0/32 (0%)**
Cost	retrieval use / calls	12/32 (38%)
	tokens / cost	13/32 (41%)
	latency	13/32 (41%)

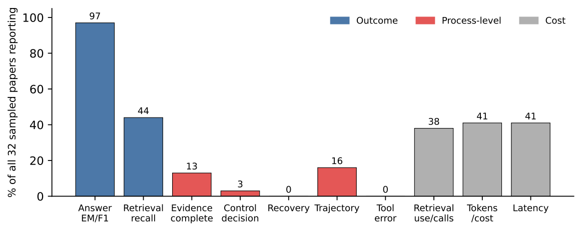

Figure 1: Percentage of the 32 systems reporting each signal. Outcome metrics dominate; agent-quality signals (recovery, tool-error, retrieval control, evidence completeness) remain sparse.

Evidence Recall (Zhuang et al., 2025); every other retrieval-reporting system uses aggregate recall@k that credits partial support. *Retrieval-control quality* is measured by exactly one system: Adaptive-RAG's question-complexity router (Jeong et al., 2024), whose ∼54.5% routing accuracy exposes a control-policy oracle gap. This is a query-level routing decision rather than a per-hop stop test, so we use the broader control-quality label. Five papers expose trajectory quality: Think-on-Graph reports path overlap with ground-truth relation paths (Sun et al., 2024); Self-Ask (Press et al., 2023), IRCoT (Trivedi et al., 2023), ReAct (Yao et al., 2023), and MDR (Xiong et al., 2021) report quantified manual analyses. Two boundary cases sharpen the point: GraphRAG reports *no* answer accuracy at all (Edge et al., 2024) (LLM-judge win-rates on an incomparable summarization axis), and StepChain (Ni et al., 2025) leaves uncertainty-aware re-decomposition, backtracking, and sub-question robustness metrics to future work, closely matching the gaps we quantify.

Interpretation. Systems are scored on the endpoint, not the behavior. A system can obtain high answer EM while stopping at the wrong hop, failing

Table 4: Exact retrieval on all official multi-support development questions. Δ is the paired aggregate-minus-joint gap; brackets give its 95% bootstrap interval.

Dataset (n)	**agg.**	**joint**	Δ **[95% CI]**
2Wiki (12,576)	73.2	45.3	27.9 [27.5, 28.4]
MuSiQue (2,417)	65.5	34.6	30.9 [29.8, 32.0]
HotpotQA (7,405)	91.0	82.7	8.3 [7.9, 8.8]

to recover, or traversing an incorrect path. These failures need not surface under endpoint-only reporting. This is precisely the concern REALM raises for agents: without process-level metrics, agentic control is credited or blamed on outcome alone, and claims of "adaptive" or "self-corrective" behavior go unvalidated.

5 Empirical Illustration

To show that the audited gap matters, we use BGE-large-en-v1.5 (Xiao et al., 2023) on the complete official development splits of 2Wiki (Ho et al., 2020), MuSiQue (Trivedi et al., 2022), and HotpotQA (Yang et al., 2018). For each split we deduplicate the union of its provided candidate contexts, then perform exact dense ranking. We report aggregate recall@20 (mean gold-passage recall) versus joint recall@20 (all gold passages retrieved), with paired 95% bootstrap intervals (5,000 resamples). Appendix B gives the protocol and support-count breakdown.

The divergence is not an artifact of mixing two- and four-support questions. On two-support subsets, the gaps remain 21.0 points (2Wiki), 22.0 (MuSiQue), and 8.3 (HotpotQA). It then widens sharply with the required evidence set: for four-support 2Wiki questions, aggregate recall is 53.4% while joint recall is only 0.7%; for four-support MuSiQue, the corresponding values are 42.8% and 3.0%. The median *weakest-link* rank is 3 on HotpotQA but 48 on 2Wiki and 56 on MuSiQue.

A fixed two-pass feedback control (query plus top-1 passage, fused by reciprocal-rank fusion (Cormack et al., 2009)) does not repair the gap. At @20 it changes aggregate recall by -4.0, -4.0, and -0.4 points and joint recall by -2.9, -1.9, and -0.2 on 2Wiki, MuSiQue, and HotpotQA, respectively; the paired joint interval includes zero only for HotpotQA. Thus a plausible expansion can look only mildly harmful under aggregate recall while losing substantially more complete evidence sets. Both effects are invisible without the completeness signal that only 13% of the audited sample reports.

Table 5: Choosing and evaluating a retrieval strategy. The metric column lists what to report beyond answer accuracy to detect the strategy's dominant failure mode.

Task property	**Strategy**	**Report also**
single-hop, high-recall corpus	static dense / hybrid top-k	recall@k
multi-hop, bounded hops, text	iterative retrieve-reason	evidence completeness; #calls
multi-hop, entity/KB-centric	graph traversal	support recall; path correctness
open-ended, variable hops, tools	agentic planning	control/stop quality; recovery; tool errors; cost

6 A Practitioner Checklist

The audit yields actionable guidance. Table 5 ties task properties to a retrieval strategy and identifies the minimal metric set that would expose its characteristic failure mode. It shows what to report *beyond* answer accuracy.

7 Related Work

Surveys of RAG and reasoning organize the field by *method* (Gao et al., 2023; Li et al., 2025; Deng et al., 2026). We are complementary and narrower: a quantitative audit of *evaluation practice* for multi-hop RAG, in the spirit of reproducibility and reporting-standard studies. To our knowledge, prior surveys have not quantified process-level metric reporting across iterative, graph, and agentic multi-hop RAG with a paper-level coding matrix.

8 Limitations

The purposive, citation-skewed corpus yields descriptive rather than population estimates, and binary codes compress partial reporting. Seven-paper repeat coding supports pooled but not stable dimension-wise κ. The illustration uses one encoder, split-induced corpora, and fixed feedback, so it diagnoses metrics rather than architectures. Logging actions is inexpensive, but judging control or recovery correctness requires task-specific supervision.

Use of AI

Generative AI assisted literature retrieval and drafting; the authors remain responsible.

References

Gordon V. Cormack, Charles L. A. Clarke, and Stefan Buettcher. 2009. Reciprocal rank fusion outperforms condorcet and individual rank learning methods. In *Proceedings of the 32nd International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 758–759.

Jingwen Deng, Jihao Huang, Zhen Hao Wong, Hao Liang, Quanqing Xu, Bin Cui, and Wentao Zhang. 2026. Data-centric perspectives on agentic retrieval-augmented generation: A survey. In *Findings of the Association for Computational Linguistics: ACL 2026*, pages 1570–1588, San Diego, California, United States. Association for Computational Linguistics.

Darren Edge, Ha Trinh, Newman Cheng, Joshua Bradley, Alex Chao, Apurva Mody, Steven Truitt, and Jonathan Larson. 2024. From local to global: A graph RAG approach to query-focused summarization. *arXiv preprint arXiv:2404.16130*.

Yunfan Gao, Yun Xiong, Xinyu Gao, Kangxiang Jia, Jinliu Pan, Yuxi Bi, Yi Dai, Jiawei Sun, Meng Wang, and Haofen Wang. 2023. Retrieval-augmented generation for large language models: A survey. *arXiv preprint arXiv:2312.10997*.

Bernal Jiménez Gutiérrez, Yiheng Shu, Yu Gu, Michihiro Yasunaga, and Yu Su. 2024. HippoRAG: Neurobiologically inspired long-term memory for large language models. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Xanh Ho, Anh-Khoa Duong Nguyen, Saku Sugawara, and Akiko Aizawa. 2020. Constructing a multi-hop QA dataset for comprehensive evaluation of reasoning steps. In *Proceedings of the 28th International Conference on Computational Linguistics*, pages 6609–6625.

Abhinav Java, Srivathsan Koundinyan, Nagarajan Natarajan, and Amit Sharma. 2025. FrugalRAG: Less is more in RL finetuning for multi-hop question answering. *arXiv preprint arXiv:2507.07634*.

Soyeong Jeong, Jinheon Baek, Sukmin Cho, Sung Ju Hwang, and Jong C. Park. 2024. Adaptive-RAG: Learning to adapt retrieval-augmented large language models through question complexity. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, pages 7036–7050.

Yangning Li, Weizhi Zhang, Yuyao Yang, Wei-Chieh Huang, Yaozu Wu, Junyu Luo, Yuanchen Bei, Henry Peng Zou, Xiao Luo, Yusheng Zhao, Chunkit Chan, Yankai Chen, Zhongfen Deng, Yinghui Li, Hai-Tao Zheng, Dongyuan Li, Renhe Jiang, Ming Zhang, Yangqiu Song, and Philip S. Yu. 2025. A survey of RAG-reasoning systems in large language models. In *Findings of the Association for Computational Linguistics: EMNLP 2025*, pages 12120–12145, Suzhou, China. Association for Computational Linguistics.

Tengjun Ni, Xin Yuan, Shenghong Li, Kai Wu, Ren Ping Liu, Wei Ni, and Wenjie Zhang. 2025. StepChain GraphRAG: Reasoning over knowledge graphs for multi-hop question answering. *arXiv preprint arXiv:2510.02827*.

Ofir Press, Muru Zhang, Sewon Min, Ludwig Schmidt, Noah A. Smith, and Mike Lewis. 2023. Measuring and narrowing the compositionality gap in language models. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 5687–5711.

Timo Schick, Jane Dwivedi-Yu, Roberto Dessì, Roberta Raileanu, Maria Lomeli, Luke Zettlemoyer, Nicola Cancedda, and Thomas Scialom. 2023. Toolformer: Language models can teach themselves to use tools. *arXiv preprint arXiv:2302.04761*.

Jiashuo Sun, Chengjin Xu, Lumingyuan Tang, Saizhuo Wang, Chen Lin, Yeyun Gong, Lionel M. Ni, Heung-Yeung Shum, and Jian Guo. 2024. Think-on-graph: Deep and responsible reasoning of large language model on knowledge graph. In *International Conference on Learning Representations (ICLR)*.

Harsh Trivedi, Niranjan Balasubramanian, Tushar Khot, and Ashish Sabharwal. 2022. MuSiQue: Multi-hop questions via single-hop question composition. *Transactions of the Association for Computational Linguistics*, 10:539–554.

Harsh Trivedi, Niranjan Balasubramanian, Tushar Khot, and Ashish Sabharwal. 2023. Interleaving retrieval with chain-of-thought reasoning for knowledge-intensive multi-step questions. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 10014–10037.

Shitao Xiao, Zheng Liu, Peitian Zhang, Niklas Muennighoff, Defu Lian, and Jian-Yun Nie. 2023. C-Pack: Packaged resources to advance general chinese embedding. *arXiv preprint arXiv:2309.07597*.

Wenhan Xiong, Xiang Lorraine Li, Srinivasan Iyer, Jingfei Du, Patrick Lewis, William Yang Wang, Yashar Mehdad, Wen-tau Yih, Sebastian Riedel, Douwe Kiela, and Barlas Oğuz. 2021. Answering complex open-domain questions with multi-hop dense retrieval. In *International Conference on Learning Representations (ICLR)*.

Zhilin Yang, Peng Qi, Saizheng Zhang, Yoshua Bengio, William W. Cohen, Ruslan Salakhutdinov, and Christopher D. Manning. 2018. HotpotQA: A dataset for diverse, explainable multi-hop question answering. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, pages 2369–2380.

Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. 2023. ReAct: Synergizing reasoning and acting in language models. In *International Conference on Learning Representations (ICLR)*.

Jiahao Zhang, Haiyang Zhang, Dongmei Zhang, Yong Liu, and Shen Huang. 2024. End-to-end beam retrieval for multi-hop question answering. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, pages 1718–1731.

Luyao Zhuang, Shengyuan Chen, Yilin Xiao, Huachi Zhou, Yujing Zhang, Hao Chen, Qinggang Zhang, and Xiao Huang. 2025. LinearRAG: Linear graph retrieval augmented generation on large-scale corpora. *arXiv preprint arXiv:2510.10114*.

A Corpus and Coding Sheet

Table 6 reports all 32 systems and all 10 reconciled codes. Paper names link to the source used for coding. Families are I (iterative), T (tool-using agent), G (graph), and D (iterative dense retrieval). Within each bit string, positions follow Table 1; 1 means reported and 0 means not reported.

Table 6: Complete paper-level coding matrix. S={S1,S2}, A={A1,...,A5}, and C={C1,C2,C3}. Each linked system name opens the source version used for coding.

System	F	S	A	C
Self-Ask	I	10	00010	010
IRCoT	I	11	00010	000
DSP	I	10	00000	000
Least-to-Most	I	10	00000	000
ITER-RETGEN	I	11	00000	110
FLARE	I	10	00000	100
Self-RAG	I	10	00000	100
Adaptive-RAG	I	10	01000	101
RaDeR	I	11	00000	000
Search-o1	I	10	00000	000
ReAct	T	10	00010	000
Toolformer	T	10	00000	100
FrugalRAG	T	11	00000	111
SCMRAG	T	10	00000	000
A-RAG	T	10	00000	010
GraphRAG	G	00	00000	010
HippoRAG	G	11	10000	011
HippoRAG2	G	11	00000	011
GNN-RAG	G	11	00000	110
Think-on-Graph	G	10	00010	100
ToG-2	G	10	00000	101
KG2RAG	G	11	00000	111
Clue-RAG	G	10	00000	110
HiRAG	G	10	00000	011
SimGRAG	G	11	00000	001
UniKGQA	G	11	00000	000
StepChain	G	10	00000	001
LinearRAG	G	11	10000	011
HopRAG	G	11	00000	111
SiReRAG	G	10	00000	001
MDR	D	11	10010	001
Beam-Retrieval	D	11	10000	000

Table 7: Paired repeat-coding matrix in S1,S2,A1–A5,C1–C3 order. The seeded, family-stratified sample has 68/70 matching cells; the two changes shown in bold were resolved from the cited sources before the corpus totals were computed.

System	Family	Initial pass	Verification pass
FLARE	iterative	1000000100	1000000100
IRCoT	iterative	1100010000	1100010000
Toolformer	tool-agent	1000000100	1000000100
SimGRAG	graph	1100000001	1100000001
HippoRAG2	graph	110000001**0**	110000001**1**
GNN-RAG	graph	1100000110	1100000110
MDR	dense	1110**0**00001	1110**1**00001

B Retrieval Illustration Protocol

We use the official 2Wiki development split (12,576 questions), MuSiQue answerable development split (2,417), and HotpotQA distractor validation split (7,405). Every question has at least two mapped support passages. The split-induced corpora are the title–text-deduplicated unions of provided candidate contexts: 56,661, 21,099, and 66,581 passages, respectively. Source revisions and SHA-256 checksums are asserted before evaluation.

BGE-large-en-v1.5 is pinned to model revision `d4aa6901...e09`; queries use its documented retrieval instruction. Normalized vectors are ranked by exact inner product. The fixed feedback query is the normalized sum of the original query and its top-1 passage; two depth-100 rankings are fused by RRF ($k_0 = 60$). We evaluate @10 and @20, use 5,000 paired bootstrap resamples (seed 20260803), and apply exact paired McNemar tests to joint-recall changes. Weakest-link ranks are exact to depth 2,000 and right-censored thereafter. We stored one record per query and independently reconstructed every point estimate and paired contingency cell.

Table 8: Single-shot recall@20 stratified by required support count. The aggregate metric increasingly masks incomplete evidence sets as the number of required passages grows.

Dataset	**Supports**	n	**Aggregate**	**Joint**
2Wiki	2	9,825	78.7	57.7
2Wiki	4	2,751	53.4	0.7
MuSiQue	2	1,252	73.2	51.2
MuSiQue	3	760	64.9	24.2
MuSiQue	4	405	42.8	3.0
HotpotQA	2	7,405	91.0	82.7